

Adetayo Olufemi Okunoye

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A results-driven professional with over seven years of experience in system engineering, data management, and state-of-the-art machine learning technologies. I possess a solid foundation in computer science, underpinned by deep expertise in algorithms, data structures, and scalable system design. My diverse background spans the development of innovative solutions, optimization of complex infrastructures, and leveraging AI to tackle challenging technical problems. With a proven track record of adaptability and impact, I am committed to advancing technology and driving excellence in software engineering and systems infrastructure.

Education History

- ❖ Ph. D in Computer Science (On Going): University of Georgia at Athens, USA. (Jan 2021, till date)
- ❖ Master's degree Computer science: Southern University and A&M Col at LA, USA. (2017 - 2019)
- ❖ Bachelor's degree Computer Eng: Ladoke Akintola University of Tech Oyo, NG. (2005 -2010)

Awards/Certifications Obtained

- ❖ Awarded Microsoft Certified Professional (MCP) – Programming with C# (June 2017)
- ❖ Best Graduate Postal Presentation (March 16, 2019) 93rd Conference Louisiana Academy of Sciences (LAS), Baton Rouge LA, USA.
- ❖ Graduate Student Scholarship award (August 2018) Southern University and A&M College Baton Rouge LA, USA.
- ❖ UGA Department Scholarship (September 2024), University of Georgia, Athens GA, USA.
- ❖ University of Georgia Graduate Teaching Assistantship (January 2021 – December 2024)

Skills

- ❖ C/C++, Java, Python, C#.NET, MVC Frameworks, Java, WCF, Web Services, Web API, Angular, MySQL, WPF, Postman, WinForms, Microsoft Visual Studio, Entity Frameworks, NHibernate, Matlab, Azure Cloud, JavaScript, TypeScript, jQuery, LINQ, XML, SQL Server, SQL Server Reporting Services, MySQL, T-SQL, Oracle.
- ❖ *Artificial Intelligence*: PyTorch, TensorFlow, keras, and scikit-learn, large language models,
 - Convolution Neural Network

Research & Publication, Software Projects

- ❖ Automated Analysis of Microscopy Images using Deep Convolutional Neural Networks – presented and published at SPIE Conference.
- ❖ Hybridization of Simulated Annealing and genetic algorithm to decongest traffic intersection in crossroad intersections
- ❖ Temperature forecasting in a hyper local weather station using Machine learning algorithms.
- ❖ Performance and analysis of Tree based model forecasting of covid-19 pandemic data

- ❖ *Wadeable Streams Project*, WadeDev is a software project dedicated to the ecological monitoring and management of wadeable streams in United states at U.S department of Interior.
- ❖ *Asset Management System*: The Asset Management System is specialized software designed to efficiently manage assets.
- ❖ *EnPower System*: The EnPower System is a comprehensive and large-scale application tailored for Nigerian Pension Administration.
- ❖ *EnInvest System*: The EnInvest System is a specialized software designed to serve pensioners in Nigeria.
- ❖ *ExamPrep*: ExamPrep is a dedicated educational software created to support students preparing for the West African Examinations Council (WAEC) and National Examinations Council (NECO) exams.
- ❖ *Laundry Kiosk System*: The Laundry Kiosk System is a digital solution designed to streamline and enhance laundry services.
- ❖ *School Manager*: School Management Systems are comprehensive software solutions tailored for educational institutions.

Employment History

- **Software Developer Intern (May 2025 – August 2025): Meta, Menlo Park, CA**
 - ❖ Led deployment of Ahead-of-Time Inductor (AOTI) optimizations for Meta's Verification Service, achieving 2–4x inference speedups across large-scale video and audio understanding models that power core social media features.
 - ❖ Designed and implemented a scalable benchmarking environment in production (Tupperware fleet + shadow traffic routing), enabling systematic performance comparisons of PyTorch compiler backends (TorchScript, TorchInductor, TorchDynamo, and AOTI).
 - ❖ Developed monitoring and profiling tools to capture latency, throughput, and memory metrics, providing data-driven insights into model performance at Meta's massive content verification scale.
 - ❖ Extended AOTI lowering pipelines for multiple media models (VCE, ACE, and Melody), ensuring performance improvements for video, audio, and music understanding tasks that directly enhance content safety and user experience.
 - ❖ Collaborated with cross-functional teams and the PyTorch open-source community to resolve complex model loading and C++ integration challenges, demonstrating adaptability, independent problem-solving, and deep technical expertise in machine learning infrastructure.
- **Graduate Teaching Assistant (Jun 2023 – May 2025): University of Georgia, Athens, GA**

Responsibilities:

 - ❖ *Assist in Course Instruction*: Support course instructors by assisting with teaching materials, facilitating lab sessions, and helping students understand complex topics.
 - ❖ *Mentor Undergraduate Students*: Provide mentorship and guidance to undergraduate students to help them grasp fundamental programming concepts and improve their coding skills.
 - ❖ *Evaluate Student Performance*: Assess student assignments and exams, offering constructive

Environment: Python, Java, C++, SQL Server.

▪ ***Software Engineer Intern/RA (May 2021-Jun 2023): U.S Department of Interior Athens GA***

Responsibilities:

- ❖ Collaborate with the development team to design, code, and implement software solutions aimed at facilitating the management and analysis of natural resource data for environmental conservation initiatives.
- ❖ Participate in the development and testing of software applications, ensuring they align with project requirements and standards.
- ❖ Support the maintenance and enhancement of existing software systems, contributing to their reliability and functionality.

Environment: ASP.NET MVC 3.0, C#, NHibernate, .Net Web API, Postman

▪ ***Software Programmer (Jan 2020 – Dec 2020): Remote Tiger, inc Greenbelt, MD***

Responsibilities:

- ❖ Collaborate with a development team to understand functional requirements for web and Windows applications, ensuring clear comprehension of project objectives.
- ❖ Convert functional specifications into working N-Tier Enterprise .NET Applications by utilizing the C# programming language and integrating with back-end services and SQL Server databases.
- ❖ Translate mockups and workflow charts into functional software solutions, facilitating user-friendly interfaces and seamless functionality for end-users.

Environment: ASP.NET MVC 3.0, C#, ADO.NET, VB .NET, .Net Web API, Azure, JavaScript, JQuery, XML, HTML5, CSS.

▪ ***Research Assistant (Aug'2017 – Dec'2019): Southern University and A&M Col (Baton Rouge, LA)***

- ❖ Direct Undergraduate Research Assistants Teaching Programming Languages (C#, Java)
- ❖ Responsible for gathering research
- ❖ Analyze research findings
- ❖ Prepare reports and paperwork related to project findings
- ❖ Maintain research databases using MS SQL Server

Environment: Python, TensorFlow, .Net Framework 4.5, MS SQL SERVER 2012, CSS, C#.Net, Visual Studio 2013

▪ ***C# .NET Developer (Apr'2016 – Aug'2017): Hartford Green Consulting – Lagos, Nigeria***

- ❖ Responsible for providing functional, technical and test specifications & requirement traceability matrix (RTM).
- ❖ Design & development of Web application using ASP.Net, AJAX, JavaScript, jQuery, CSS, WCF, XML, technologies
- ❖ Service oriented architecture using Web API
- ❖ Followed TDD and unit testing, used Moq framework for mocking and breaking dependencies.
- ❖ UI design and end to end UI development with backend using ASP.NET MVC, HTML5, CSS3, Bootstrap, JavaScript, Angular JS.
- ❖ Used PowerShell for scripting batch processes and creating customized system management tools.

- ❖ Used CORS to allow cross domain access between API and MVC application.
- ❖ Design and deployment of project using Microsoft Azure.
- ❖ Data layer using Entity Framework 6.1 over SQL Server database and LINQ for generating queries
- ❖ Used N-tier architecture for presentation layer, the Business and Data Access Layers and were Coded using C# also worked with agile methodology for Application Development.
- ❖ Converting .Net application to Microsoft Azure Cloud Service Project as part of cloud deployment.
- ❖ Migrating SQL Server database to Windows Azure SQL Database and updating the Connection Strings based on this
- ❖ Created several Complex Reports (sub reports, graphical, multiple groupings, drilldowns, parameter driven, formulas, summarized and conditional formatting) using SSRS.

Environment: .Net Framework 4.5, WinForm, Window Presentation Foundation (WPF) UI Framework, .Net Web API, C#, ASP.NET MVC, ADO.NET, AJAX, JavaScript, JQuery, CSS, Web API, XML, SQL Server, Visual Studio 2013.

▪ **Full Stack .Net Developer (Feb'2015 – March'2016): xPlug Technologies - Lagos, Nigeria**

- ❖ Designed and developed intranet web applications using the ASP.NET MVC framework and integrated additional functionality using C++.
- ❖ Followed the Scrum Agile Methodology during project development to ensure efficient collaboration and timely delivery.
- ❖ Utilized the MVC 5.0 Razor-View Engine for reusing individual user controls without coupling them to underlying data elements.
- ❖ Leveraged both client-side and server-side validation, better session management, and powerful data access management using ASP.NET MVC (5.2).
- ❖ Designed and coded Win Forms and implemented logic using both C#.Net and C++ to ensure robust application performance.
- ❖ Configured trust relationships between ADFS servers and relying parties, including the setup of relying party applications for the ADFS server.
- ❖ Deployed services across environments using both Node Server and IIS server, tailoring solutions to client requirements.
- ❖ Developed responsive web designs using Bootstrap for enhanced user experience on multiple devices.
- ❖ Extensively created and deployed WCF services, including the implementation of bindings, endpoints, and service contracts.
- ❖ Implemented database connectivity using ADO.NET, creating stored procedures and triggers for automated operations and relational updates.
- ❖ Employed LINQ for efficient querying of datasets and utilized C++ for handling performance critical tasks in backend processes.
- ❖ Adopted Test-Driven Development (TDD) practices, ensuring code reliability and prepared comprehensive documentation for future reference and upgrades.

Environment: WPF, ASP.NET, Visual Studio 2012, MVC 5.0, LINQ, SQL Server 2012, IIS 7.0, TFS, WINDOWS SERVICES, C#.NET, C++, .NET Framework 4.6.1, AJAX, CSS, jQuery, JavaScript

▪ **Software Developer (Jan'2012 - Feb'2015): TOPGY Systems and Management Consulting - Lagos, Nigeria**

- ❖ Gathered Client requirements and converting them into the Technical Specifications and developed Web forms using C#.NET
- ❖ Developed ASP.NET MVC4 application in Test Driven Development environment using Microsoft Test as the Testing framework.
- ❖ Used Microsoft Entity Framework Code First approach for the Data Access Layer for the ASP.NET MVC4 application.
- ❖ Worked with Language Integrated Query (LINQ) to query the data against the database (LINQ to SQL), dataset (LINQ to dataset), XML (LINQ to XML).
- ❖ Provided production support on the existing Power BI Dashboards and other ETL processes to resolve the issues quickly and published Power BI Desktop reports created in Report View to the Power BI service
- ❖ Scheduled maintenance Jobs and Alerts using SQL Server Agent and creating jobs on all the MS SQL servers using Multi Server Administration.
- ❖ Used ADO.NET objects like Data Reader, Data Adapter and Dataset, Data Grid for data access.
- ❖ Successfully tested and implemented Log Shipping in MS SQL Server 2008/2005/2000 and used as backup solution in case disaster recovery.
- ❖ Have used a variety of SCM tools for collaborative development: Git, Subversion, Clear Quest, TFS, SharePoint, and Confluence
- ❖ Knowledge in syncing On-premises Windows Server Active Directory to Azure AD(AAD) using Azure AD connect
- ❖ Implemented changes in coding and designs using knowledge of VB .NET.
- ❖ Extract and analyze data using Power Query, PivotTables, MS Excel, POWER BI and SSAS ➦ Building and Installing servers through Azure Resource Manager Templates or Azure Portal ➦ Bi-weekly code reviews, deployments and maintenance.

Environment: ASP.NET MVC 3.0, C#, ADO.NET, VB .NET, Visual Studio 2010, SQL Server 2008 R2, Angular JS, MVVM, TFS 2010, Azure, JavaScript, JQuery, XML,HTML5, CSS, NUnit, Selenium.

▪ **Mandatory NYSC (Jan'2011 – Dec'2011): Saint Cyprian Special Science School – Enugu, Nigeria**

- ❖ Design and implement engaging lesson plans aligned with educational standards, covering fundamental IT concepts such as programming, networking, and cybersecurity.
- ❖ Instruct students in using computer software, hardware, and emerging technologies, ensuring they develop practical skills for academic and professional use.
- ❖ Organize and supervise practical sessions, projects, and assignments to enhance students' understanding of IT applications in real-world scenarios.
- ❖ Evaluate student performance through quizzes, tests, and projects, providing constructive feedback to help them improve and master IT concepts.
- ❖ Continuously update knowledge of the latest technological advancements and incorporate innovative tools and techniques into the classroom to keep students engaged and prepared for future challenges.

References

† Available Upon Request